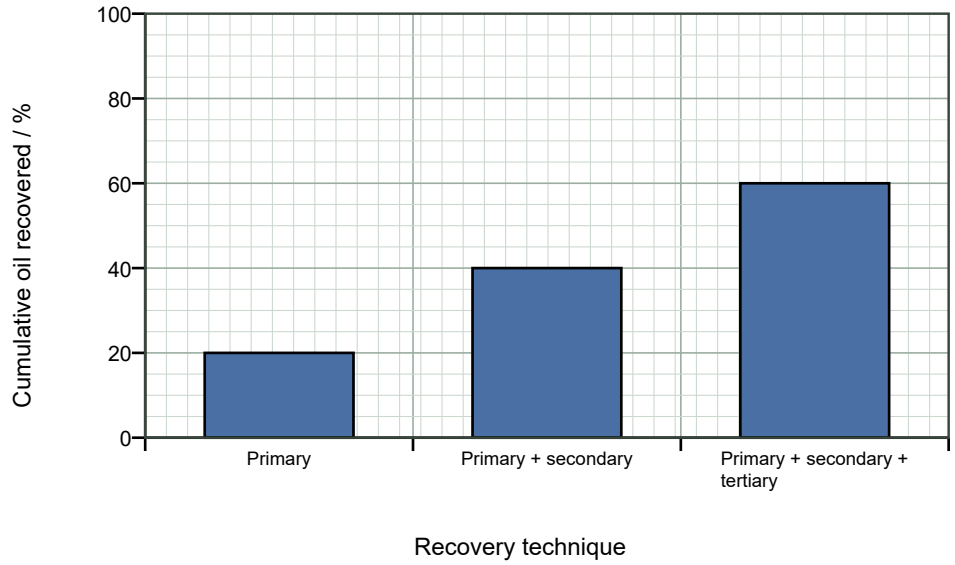
Questions by topic: Fossil fuels & nuclear power: sustainability and new technologies - Set 1 (70 marks)

Answer all questions in the spaces provided.

0 1 Figure 1 shows the percentage of the oil in a reservoir that can be recovered using different recovery techniques.

Figure 1



0 1 . 1 Describe how primary oil recovery works and explain why it recovers only about 20% of the oil.

[2 marks]

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0 1 . 2 Describe secondary recovery and explain how it can be linked to carbon capture and storage.

[2 marks]

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0 1 . 3 Describe **three** tertiary (enhanced) recovery techniques.

[3 marks]

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0 1 . 4 Explain **three** advantages of directional drilling.

[3 marks]

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0 2 . 1 Explain why oil and gas trapped in shale cannot be extracted by conventional drilling and describe how hydraulic fracturing (fracking) allows extraction.

[3 marks]

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0 2 . 2 Describe **four** environmental concerns about fracking and suggest how they may be reduced.

[5 marks]

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0 2 . 3 Explain why tar sands and oil shales are not included in most estimates of oil reserves despite containing more oil than conventional crude oil.

[2 marks]

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0 3 . 1 Describe coal gasification and coal liquefaction and explain how each increases the usefulness of coal.

[4 marks]

Do not write outside the box

0 3 . 2 Describe what methane hydrate is and where it is found.

[2 marks]

0 3 . 3 Describe **two** proposed methods of extracting methane from hydrate deposits and explain **two** environmental risks.

[4 marks]

10

0 4 Table 1 compares a 1000 MW coal-fired power station with a 1000 MW nuclear power station.

Table 1

Feature	Coal-fired	Nuclear (uranium)
Fuel used per year	3.5 million tonnes	30 tonnes
CO_2 emitted per kWh / g	900	12 (mostly fuel processing and construction)
Construction cost / £ billion	2	20
Fuel cost per kWh / pence	3.0	0.5
Typical response to changes in demand	Hours	Slow (days)

0 4 . 1 Explain how nuclear fission releases energy in a reactor.

[3 marks]

Do not write outside the box

0 4 . 2 Use Table 1 to explain why nuclear power is used for base-load electricity but not for meeting peaks in demand.

[2 marks]

0 4 . 3 Use Table 1 to explain why nuclear power stations can be built where coal-fired stations could not.

[2 marks]

0 4 . 4 Explain why the CO_2 emissions of nuclear power are not zero.

[1 mark]

0 4 . 5 Give **two** reasons why the true cost of nuclear electricity is uncertain.

[2 marks]

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0 5 . 1 Describe **three** factors that have restricted the growth of nuclear power.

[3 marks]

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0 5 . 2 Explain how a plutonium 'breeder' reactor can produce more fissile fuel than it uses.

[3 marks]

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0 5 . 3 Give **two** advantages and **two** disadvantages of thorium reactors compared with uranium reactors.

[4 marks]

Advantages .....

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Disadvantages .....

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0 6 . 1 Explain the difference between nuclear fission and nuclear fusion.

[2 marks]

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0 6 . 2 State the sources of the two fuels used in fusion research.

[2 marks]

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0 6 . 3 Explain why very high temperatures, a vacuum and a magnetic field are needed in a toroidal (torus) fusion reactor.

[3 marks]

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0 6 . 4 Give **two** potential advantages of fusion over fission and one reason why it is not yet a commercial energy source.

[3 marks]

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0 7 . 1 Evaluate the extent to which new technologies could allow fossil fuels and nuclear fission to continue to supply a large proportion of the world's energy sustainably.

[9 marks]

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Extra space

0 7 . 2 State the approximate percentage of natural gas in a reservoir that can be recovered using its own pressure.

[1 mark]

10

END OF QUESTIONS